

## Solution to the Given Logarithmic Equation

Given equation:

$$\log_4(6 - x) = \log_2 x$$

Step 1: Convert the logarithmic bases

We express  $\log_4$  in terms of base 2:

$$\log_4(6 - x) = \log_2(6 - x) / \log_2 4$$

Since  $\log_2 4 = 2$ , we get:

$$(\log_2(6 - x)) / 2 = \log_2 x$$

Step 2: Eliminate the fraction

Multiplying both sides by 2:

$$\log_2(6 - x) = 2 \log_2 x$$

Step 3: Apply logarithmic identity

Using the identity  $a \log_b c = \log_b c^a$ , we rewrite:

$$\log_2(6 - x) = \log_2 x^2$$

Since  $\log_2 A = \log_2 B$  implies  $A = B$ , we get:

$$6 - x = x^2$$

Step 4: Solve the quadratic equation

Rearrange:

$$x^2 + x - 6 = 0$$

Factorize:

$$(x + 3)(x - 2) = 0$$

Thus, the possible values of  $x$  are:

$$x = -3 \text{ or } x = 2$$

Since we need a positive  $x$ , we take:

$$x = 2$$

Step 5: Evaluate  $\log_4 (8x)$

We substitute  $x = 2$ :

$$\log_4 (8 * 2) = \log_4 (16)$$

Since  $16 = 4^2$ , we get:

$$\log_4 16 = 2$$

Final Answer:

$$x = 2, \log_4 (8x) = 2$$